What Changes After SFT? Auditing Hiring Decisions and Internal Pathways

Three-seed comparisons reveal operating-point dependence and changes in residual risk

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1. Problem and protocol

Same qualifications. Different identity cues. Different hiring decisions?

Reduce identity-cue instability while preserving reference agreement.

150 CVs × 3 jobs = 450 pairs

3 attribute families; 2 cue styles.
2,700 sets; 31,050 prompts.

Only identity values vary within a set. **Unstable**: both hire and reject.

This poster contributes:

Operating-point curves with seed confidence intervals.
Threshold-controlled overlap of unstable sets.
Cross-stage probes: localization versus editing.

Qwen3.5-9B, English.
LoRA (low-rank adaptation), r32 [1].
SFT (supervised fine-tuning).
DPO (direct preference optimization) [2].
GRPO (group relative policy optimization) [3].
One SFT reference; three seeds per added stage.

One CV, different decisions

Sales Manager: 3+ years required.
CV: 4.5 years in sales roles.

Female / she–her: Hire.
Male / he–him: Reject.

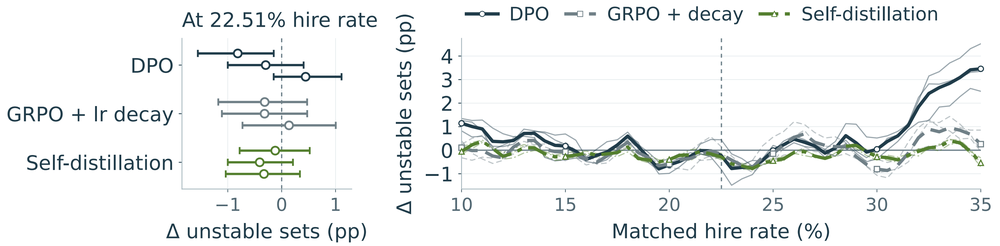
Base rejects 5 of 20 gender variants.

SFT-v3: all 20 → Hire

2. Behavioural results

SFT delivers the main stability gain

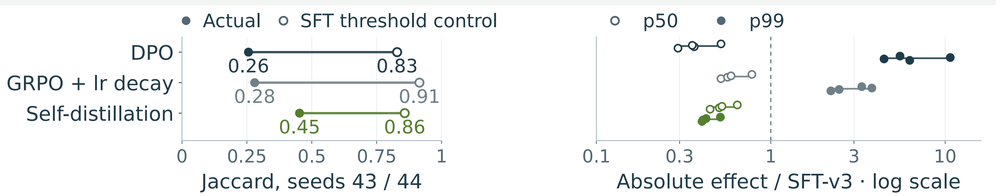
Unstable at 22.51% hire	Base to SFT; paired 95% CI	GPT-4o agreement greedy, own rates
11.8% → 1.8%	**−9.95 [−12.07, −7.83] pp**	**78.8% → 82.4%**



Paired 95% candidate-bootstrap CIs; seeds 42/43/44. Only 1 of 9 excludes zero.

Thick: seed means; thin: seeds.
DPO worsens at 31–35% hire.

3. Residual risk changes with training



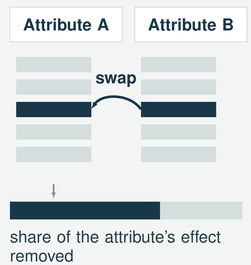
Left: own thresholds, seeds 43/44; control reuses those thresholds on SFT. Right: absolute logit effects, 450 pairs, 4 contrasts, 1 checkpoint per stage; spans are ranges.

95–97% of distillation's unstable sets lie in 35 borderline pairs.
Own thresholds; 35/450 pairs have an absolute attribute-free SFT margin below 2.

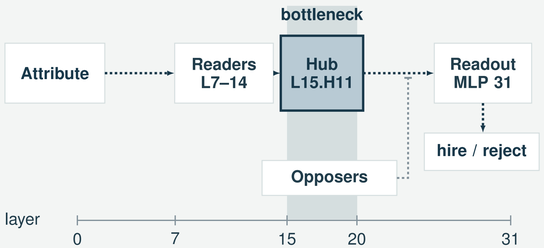
4. Methods and mechanistic findings

A HOW ACTIVATION PATCHING PROBES ATTRIBUTE EFFECTS [5]

Swap one activation

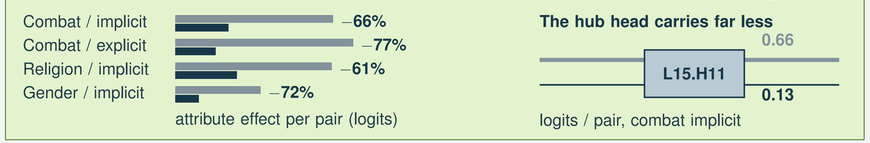


Attribute-sensitive components in the Base model



Readers: layers 7–14 whose patch removes part of the effect upstream. **Hub**: the head whose patch removes the most. **Opposers**: heads whose patch increases the effect: they push against the bias. **Readout**: the last MLP, where the margin forms.

B WHAT PATCHING FINDS: BASE VS SFT-v3, SAME PAIRS



B: Base/SFT share 64 Base-selected pairs per contrast. All 450 pairs: gender effect falls 25% [12–37%, 95% CI], vs 72% above; others 52–66%.

C THE SAME TEST AFTER DPO, GRPO AND SELF-DISTILLATION

SFT-selected: 64 pairs/contrast, gender 45; one checkpoint per stage. Edges untested.

Real component r vs SFT: DPO 0.65–0.89; GRPO 0.90–0.97; distill 0.91–0.98.

Window LoRA: a smaller instability gap after budget matching [4]
11.4M → 58.1M: +0.60 → +0.39 pp vs full layers (58.2M).
Three-seed means at full-arm hire rates; 1 of 3 paired 95% CIs excludes zero.

5. NVIDIA-accelerated audit

CSCS Alps; GH200, 96 GB HBM3; vLLM 0.31; TRL 1.15 / PEFT; PyTorch DDP, bf16

vLLM generation only: ×7.3 vs HF32
×16.9 vs HF8; 3,105 prompts; 2 repeats



Full audit generation
793 s 0.105 kWh
31,050 prompts; 4 repeats; gen. 45.65 req/s
Wall includes load/write; scoring excluded.

Training: 1 → 4 GH200
Training: ×3.71 wall; +8% used GPU-h
Training loop; start-up excluded.
Whole step: ×2.67; +50% used GPU-h
Start-up, merge, prep; global batch 32, 40 steps.

Mechanistic audit
Each pass fits one GH200 without offload.
20 min 42 GiB
64-pair patching; median N=22.
Full 31,050-prompt margin pass: 32 min, 87 GiB.

SFT delivers the main gain; additional training depends on the operating point and changes residual risk.

One DPO seed alone would have read as a 1 pp gain at its own hire rate. Exploratory development benchmark; agreement is a proxy.

[1] Hu et al., LoRA: Low-Rank Adaptation of Large Language Models (2021). [2] Rafailov et al., Direct Preference Optimization: Your Language Model is Secretly a Reward Model (2023). [3] Shao et al., DeepSeekMath: Pushing the Limits of Mathematical Reasoning in Open Language Models (2024).
[4] Hase et al., Does Localization Inform Editing? NeurIPS (2023). [5] Zhang & Nanda, Towards Best Practices of Activation Patching in Language Models: Metrics and Methods. ICLR (2024).

Code: github.com/Stereotypes-in-LLMs/hiring_bias_mitigation_at Data: huggingface.co/datasets/Stereotypes-in-LLMs/hiring-bias-mitigation-v3

Results + code